

9. **SCIENTIFIC INQUIRY** Scientists studying scrub jays found that “helpers” often assist mated pairs of birds by gathering food for their offspring. (a) Propose a hypothesis to explain what advantage there might be for the helpers to engage in this behavior instead of seeking their own territories and mates. (b) Explain how you would test your hypothesis. If it is correct, what results would you expect your tests to yield?
10. **SCIENCE, TECHNOLOGY, AND SOCIETY** Researchers are very interested in studying identical twins separated at birth and raised apart. So far, the data reveal that such twins frequently have similar personalities, mannerisms, habits, and interests. What general question do you think researchers hope to answer by studying such twins? Why do identical twins make good subjects for this research? What are the potential pitfalls of this research? What abuses might occur if the studies are not evaluated critically? Explain your thinking.
11. **WRITE ABOUT A THEME: INFORMATION** Learning is defined as a change in behavior as a result of experience. In a short essay (100–150 words), describe how heritable information contributes to the acquisition of learning, using some examples from imprinting and associative learning.

12. **SYNTHESIZE YOUR KNOWLEDGE**



Acorn woodpeckers (*Melanerpes formicivorus*) stash acorns in storage holes they drill in trees. When these woodpeckers breed, the offspring from previous years often help with parental duties. Activities of these nonbreeding helpers include incubating eggs and defending stashed acorns. Propose some questions that a behavioral biologist could ask about the proximate and ultimate causation of these behaviors.

For selected answers, see Appendix A.